MATH 3551 A — Abstract Algebra I
Practice Midterm 2 — Version 2

Taylor Dupuy, University of Vermont • Fall 2026

For practice — Midterm 2 is Friday, November 6, 2026 • Time limit: 50 minutes • Total: 100 points

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NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — "by Lagrange's Theorem, $|H|$ divides 24" earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

Version 2 — September 28, 2026. Generated by GPT-5.6 Sol in the Codex desktop harness. Please notify Professor Dupuy if something looks off.

What changed from Version 1. The October 26 revision, drafted with Codex, moves §§3.4–3.5 and Chapter 4 to Midterm 3. This version now stops at §3.3. Material deferred from Midterm 1 — §1.7, §2.2, and §§3.1–3.2 — remains included. The October 20 revision shortened the exam and rescored the retained parts. Version 1 remains available on the course webpage so that the changes can be compared directly.

This practice exam targets about twenty minutes of work in a fifty-minute period. Please report any error, ambiguity, out-of-coverage question, or item that takes much longer than its point value suggests.

Problem 1. (20 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Five points each: two for the circle, three for the reason.

(a) **T** **F** If $H \leq G$ and $N \trianglelefteq G$, then HN is a subgroup of G.

(b) **T** **F** If $K \trianglelefteq H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.

(c) **T** **F** Every subgroup of index 2 in a group is normal.

(d) **T** **F** Every quotient group is abelian.

Problem 2. (30 points) **A quotient of $\mathbb{Z}/30\mathbb{Z}$.**

Work additively in $G = \mathbb{Z}/30\mathbb{Z}$. Let $H = \langle \overline{6} \rangle$ and $N = \langle \overline{10} \rangle$.

(a) (15 points) List H, N, $H \cap N$, and $H + N$.

(b) (15 points) Identify the two groups in the Second Isomorphism Theorem $H/(H \cap N) \cong (H + N)/N$ and verify the isomorphism in this example.

Problem 3. (30 points) **Subgroups above N.**

Work additively in $G = \mathbb{Z}/18\mathbb{Z}$ and let $N = \langle \overline{6} \rangle$.

(a) (15 points) List every subgroup H of G that contains N, giving a generator and order for each.

(b) (15 points) Identify G/N and match each subgroup from (a) with its corresponding subgroup H/N of G/N.

Problem 4. (20 points) **A normalizer in S_3.**

Let $H = \langle (1\,2) \rangle \leq S_3$.

(a) (10 points) Determine $C_{S_3}(H)$.

(b) (10 points) Determine $N_{S_3}(H)$ and decide whether $H \trianglelefteq S_3$.

End of practice exam. Total: $20 + 30 + 30 + 20 = 100$ points.